

WAVEFRONT

A Structural Translation Engine for Discovering Transportable Truths

Research Program Proposal — Position Paper

Find the mapping. Specify the contract. Test the transfer.

Abstract

Most scientific breakthroughs involve the transport of a principle discovered in one domain into a structurally different domain. Existing AI systems are strong at detecting surface similarity but weak at identifying which properties survive transformation and which do not. This paper proposes Wavefront, a research program that makes transportability a first-class object of study. The central primitive is the Preservation Contract: a formal specification of what is preserved, what is destroyed, and with what confidence when a process from one domain is mapped onto another. We develop the theoretical foundations of this framework, propose a system architecture, define concrete evaluation metrics, and identify principal failure modes. We argue that Wavefront does not constitute a theory of everything but rather a tractable science of structural translation—one that is falsifiable, cumulative, and useful.

1. Introduction

Scientific progress frequently arrives through a particular epistemic move: a principle governing one class of phenomena turns out to govern another class that looks, on the surface, entirely different. Evolution by natural selection found expression in genetic algorithms. The immunological logic of adaptive memory found application in cybersecurity. Shannon's entropy framework, originally about communications channels, quietly absorbed statistical mechanics and emerged as a general theory of information. In each case, a successful transport occurred—and in each case, something was preserved while something else was deliberately discarded.

Current AI research has invested heavily in the detection of surface similarity. Embedding spaces, contrastive learning objectives, and retrieval-augmented systems all excel at identifying that two things are alike. They are substantially weaker at answering the more important question: which properties of the source domain remain valid predictions in the target domain, and which fail? This distinction—between resemblance and structural equivalence—is the central concern of Wavefront.

This proposal makes three claims. First, that transportability is a tractable research object: it can be formally defined, empirically studied, and incrementally improved. Second, that the Preservation Contract is a useful primitive for capturing transport claims in a way that is testable and falsifiable. Third, that a research program organized around these objects could accelerate discovery in science and engineering in ways that purely similarity-based systems cannot.

We do not claim that Wavefront will automate scientific breakthroughs. We claim that it can formalize a class of inference that humans already perform intuitively but inconsistently—and that formalization makes that class of inference auditable, improvable, and scalable.

2. Foundational Shift: From Representation to Transport

Traditional scientific modeling asks: what is the underlying structure of this phenomenon? The goal is a faithful representation of reality—a model that captures the thing.

Wavefront asks a different question: given two processes, what structural properties survive the transformation from one to the other? The goal is not a model of the thing but a model of the mapping between things.

This shift has precedents. Emmy Noether’s theorem established that every continuous symmetry of a physical law corresponds to a conservation law—a result about what survives transformation under a specified group of symmetry operations. Category theory generalized this intuition mathematically: rather than studying objects in isolation, it studies the morphisms between them and the properties those morphisms preserve. Information theory studies which properties of a signal survive encoding, compression, and transmission through a noisy channel.

Wavefront extends this orientation into a general research program. The question is not “what is this system?” but “what survives when this system is represented, abstracted, implemented, or instantiated differently?”

The central question: Given Process A and Process B, and a specified transformation T , what properties P of A remain valid in B under T , and what properties Q do not?

3. Core Concepts

3.1 Transportable Truth

A transportable truth is a statement that remains valid under a specified class of transformations. This is an operational definition, not a metaphysical claim.

Examples exist across disciplines. In physics, statements that are invariant under Lorentz transformations are transportable truths about relativistic systems. In mathematics, theorems about graph connectivity remain valid under bijective relabeling of vertices. In engineering, modularity principles discovered in hardware design transferred successfully to software architecture. In information theory, the channel capacity formula survives changes in the physical encoding medium.

The critical word is specified. A transportable truth is always relative to a class of transformations. No statement is unconditionally transportable; every transfer has scope conditions. Part of Wavefront’s task is to make those scope conditions explicit rather than implicit.

3.2 Similarity versus Structural Equivalence

The most common failure mode in analogical reasoning is conflating surface similarity with structural equivalence. Consider two frequently cited analogies:

The neuron-galaxy analogy: neurons and galaxies both have central bodies with extending tendrils. The visual resemblance is striking. But the structural properties of neurons—electrochemical signaling, threshold activation, synaptic plasticity, network topology that produces computation—do not transfer. There are no predictions about galaxies that follow from neuron dynamics. The analogy is decorative.

The immune system-cybersecurity analogy: the immune system and a distributed computing security architecture share no visual resemblance whatsoever. But both exhibit distributed sensing across a large surface area, local detection and response, adaptive memory that improves with exposure, graceful degradation when components fail, and layered redundancy. Predictions derived from immunology—for example, that memory-based recognition outperforms stateless scanning, or that autoimmune-like false positives become a design challenge at scale—have transferred successfully.

The difference is not how similar the two systems look. It is whether the constraints, dynamics, and failure modes governing one system are preserved in the other. Preservation of constraints is what makes a transfer useful. Appearance is irrelevant.

3.3 Preservation and Selective Loss

A critical insight, easily overlooked: useful abstraction requires both preservation and selective loss. These are not in tension—they are jointly necessary.

A mapping that preserves everything is not an abstraction. It is a copy. No information is gained about what matters versus what is incidental. A mapping that preserves nothing is noise. It communicates that two systems are different, which is trivially true of almost any two systems.

Meaning lives in the middle. A successful Preservation Contract identifies which structural features travel across domains—and which features are implementation-specific accidents that should be left behind. The biochemical signaling mechanisms of the immune system are not transported to cybersecurity. The distributed adaptation logic is. Knowing which is which is the substance of the transport claim.

3.4 Transformations as First-Class Objects

Most attention in cross-domain transfer goes to the endpoints: source and target. Wavefront proposes that the transformation itself deserves independent study.

Different types of transformations tend to preserve different types of properties. Abstraction tends to preserve topological and relational structure while discarding physical instantiation. Compression tends to preserve high-frequency information while discarding low-frequency detail. Hierarchical decomposition tends to preserve compositional structure while discarding inter-level coupling. Error-correcting encoding preserves the information content of a message while introducing redundancy. Feedback insertion tends to preserve stability properties while changing transient dynamics.

Discovering which transformation classes reliably preserve which property classes is as important as discovering specific invariants. A well-characterized transformation is a reusable tool. A library of transformations with known preservation profiles is infrastructure for accelerated scientific discovery.

4. The Preservation Contract

The primitive object of Wavefront is the Preservation Contract. A contract specifies:

- Source Process: the origin domain, with its states, dynamics, constraints, and failure modes
- Target Process: the destination domain, similarly characterized
- Transformation: the formally specified mapping from source to target
- Preserved Properties: the structural features that remain valid under the mapping
- Non-Preserved Properties: the features that are explicitly left behind
- Preservation Strength: a quantitative or ordinal measure of how completely each property is preserved
- Confidence: an epistemic estimate of the reliability of the contract
- Supporting Evidence: empirical, theoretical, or experimental grounding for the claims

A worked example:

Source: Biological immune system Target: Distributed cybersecurity architecture Transformation: Abstraction (remove biochemical substrate; retain coordination logic) Preserved: • Distributed sensing across exposed surface area • Local detection and autonomous response • Adaptive memory indexed by prior exposure • Graceful degradation under partial component failure • Layered redundancy Not Preserved: • Biochemical signaling mechanisms • Metabolic energy constraints • Evolutionary timescales of adaptation • Somatic hypermutation Preservation Strength: High (coordination logic), Low (adaptation speed) Confidence: 0.82 (multiple independent implementations validated) Evidence: Design pattern literature; intrusion detection system benchmarks; theoretical analysis of distributed adversarial systems

The Preservation Contract is more useful than a raw analogy because it is explicit about what fails. Most analogies are stated in terms of resemblance (“the immune system is like a security system”). Contracts are stated in terms of preservation scope, which makes them falsifiable. The claim “graceful degradation is preserved” generates testable predictions. The claim “these two things look alike” does not.

5. System Architecture

Wavefront proposes a six-layer architecture. Each layer takes structured input, produces structured output, and faces identifiable open research problems.

Layer 0 — Compression

Input: Raw system descriptions (text, code, scientific literature, simulation outputs). Output: Compact structural representations. Methods: Minimum Description Length compression, representation learning, mechanistic abstraction, graph compression. Open problems: How to separate structural properties from implementation details automatically. Failure modes: Lossy compression that discards structurally relevant features.

Layer 1 — Process Representation

Input: Compressed system descriptions. Output: Uniform process representations using a shared ontology. The ontology includes states, dynamics, constraints, resources, interfaces, feedback loops, failure modes, and observables. Open problems: Ontology stability across domains; edge cases where the ontology is underspecified. Failure modes: Category mismatch; forcing systems into ontological bins that distort their structure.

Layer 2 — Contract Discovery

Input: Pairs of process representations. Output: Candidate Preservation Contracts. Methods: Constraint matching, structural isomorphism detection, simulation-based behavioral comparison, large language model-assisted hypothesis generation. Open problems: Tractable search over exponentially large contract spaces; avoiding spurious matches. Failure modes: High false positive rate; contracts that describe visual similarity rather than structural equivalence.

Layer 3 — Constraint Verification

Input: Candidate contracts. Output: Verified or falsified contracts with confidence scores. Tests: Stability preservation, information flow preservation, conservation structure, failure mode alignment, predictive consistency across held-out cases. Open problems: Verification for continuous or high-dimensional systems; handling underspecified domains. Failure modes: Verification that is computationally tractable only for toy systems.

Layer 4 — Predictive Transport

Input: Verified contracts. Output: Falsifiable predictions about the target domain derived from source domain knowledge. Example: if mechanism X improves robustness in Domain A, and robustness is a preserved property of the $A \rightarrow B$ contract, then mechanism X is predicted to improve robustness in Domain B. Open problems: Quantifying prediction confidence; handling contracts with partial preservation. Failure modes: Overgeneralization; confident predictions in domains where preservation breaks down.

Layer 5 — Experimental Evaluation

Input: Transported predictions. Output: Empirical validation or falsification. Methods: Simulation, historical reconstruction, human expert review, controlled experiments. This layer closes the loop: failed predictions generate information about which preservation claims were incorrect, feeding back to improve Layer 3. Open problems: Ground truth is expensive to obtain. Failure modes: Retrospective rationalization of failed transfers.

6. Toward Formalization

We do not present a complete formal theory. We propose candidate formal representations and identify connections to existing mathematical frameworks.

6.1 Process

A process P is a tuple (S, D, C, R, I, F) where S is the state space, D is the dynamics (a transition function or differential equation), C is the set of constraints, R is the resource structure, I is the interface specification, and F is the set of failure modes. This representation is compatible with both discrete and continuous systems.

6.2 Transformation

A transformation $T: P_1 \rightarrow P_2$ is a morphism between process representations. In categorical terms, Wavefront studies which functors preserve which properties when processes are viewed as objects in appropriate categories. Connections to causal representation learning are direct: transformations that preserve causal structure are more likely to support successful predictive transport than transformations that merely preserve correlational structure.

6.3 Preservation Property

A preservation property ϕ is a predicate defined on process representations. We say T preserves ϕ (written $T \models \phi$) if $\phi(P_1) \Rightarrow \phi(T(P_1))$ holds over the relevant class of instances. Preservation strength is a measure of how tightly this implication holds—a probability over a reference class of instances, or a worst-case bound over a specified distribution.

6.4 Preservation Contract

A contract $K = (P_1, P_2, T, \Phi^+, \Phi^-, s, c, E)$ where Φ^+ is the set of preserved properties, Φ^- is the set of non-preserved properties, s is a preservation strength vector, c is a confidence estimate, and E is the evidence structure. A contract is well-formed if $\Phi^+ \cup \Phi^-$ covers the properties of interest and the claims are independently testable.

These representations connect naturally to dynamical systems theory (stability and robustness properties), control theory (feedback structure and constraint satisfaction), information theory (information retention under compression), and mechanistic abstraction literature in AI interpretability.

7. Research Objectives and Evaluation

7.1 Three Core Objectives

Discovery Objective: Identify equivalence classes of processes—systems that, under appropriate transformation, preserve the same structural properties. Question: which systems are secretly playing the same game?

Distinction Objective: Identify the boundaries where transport fails. Failures are not embarrassments; they are data. A contract that specifies what does not travel is as valuable as one that specifies what does.

Contract Calibration Objective: Estimate preservation strength and contract confidence. Prevent contract inflation—the tendency to overclaim the scope of a successful analogy. The goal is calibrated uncertainty, not maximal generality.

7.2 Evaluation Metrics

Preservation Accuracy: Given a contract claiming property P is preserved, is P actually preserved in the target domain? Measured against ground-truth expert assessment and experimental outcomes.

Distinction Accuracy: Given a contract claiming property Q is not preserved, is Q actually absent in the target domain? Prevents the failure mode of treating everything as preserved.

Prediction Transfer Success: Do predictions generated by transported contracts validate empirically? This is the ultimate metric—it tests whether the formalism produces useful science.

Expert Agreement: Do domain experts independently identify similar contracts for the same system pairs? High inter-rater agreement is evidence that contracts are capturing real structure rather than subjective impression.

Novel Transfer Yield: How many successful transport hypotheses does the system generate that were not previously known? This measures the generative value of the framework beyond systematizing what humans already know.

8. Historical Benchmark Dataset

Evaluation requires ground truth. We propose a benchmark dataset of historically successful and failed transfers, annotated with full Preservation Contracts.

Candidate entries for the benchmark include:

- Evolution → Genetic Algorithms: preserved selection pressure, fitness landscape dynamics, population diversity mechanisms; not preserved: molecular biology, reproductive biology
- Biological Neural Systems → Artificial Neural Networks: preserved: distributed representation, hierarchical composition, gradient-based learning; not preserved: metabolic constraints, temporal spike coding, embodiment
- Immune System → Cybersecurity: preserved: distributed sensing, adaptive memory, graceful degradation; not preserved: biochemical substrate, evolutionary timescales
- Ant Colony Dynamics → Routing Algorithms: preserved: emergent optimization, pheromone-analog gradient following, decentralized coordination; not preserved: physical embodiment, colony lifecycle
- Thermodynamics → Information Theory: preserved: entropy as measure of uncertainty, second law analog (information cannot increase without cost), equilibrium concepts; not preserved: physical substrate, temperature as physical quantity
- Failed transfer: Hydraulic models of cognition → Psychology: surface metaphor (mental ‘pressure’, ‘flow’, ‘valves’) without structural preservation; generated no successful predictions

Each entry would be annotated by domain experts with explicit preservation and non-preservation claims, confidence estimates, and evidence. Inter-rater reliability would be measured using Cohen’s kappa on property-level agreement. We anticipate substantial disagreement on preservation strength and moderate agreement on the presence or absence of specific preserved properties.

9. Failure Modes and Critiques

We take the following critiques seriously. Each represents a genuine threat to the research program.

Contracts may be irreducibly subjective

The identification of preserved properties may be a post-hoc rationalization of successful analogies rather than a predictive procedure. If any successful analogy can be redescribed as a Preservation Contract after the fact, the framework is unfalsifiable.

Response: This is the strongest critique. The response is methodological: contracts must be specified before predictions are generated, not after outcomes are observed. Pre-registration of contracts and predictions, analogous to clinical trial pre-registration, is a necessary component of the evaluation framework. Retrospective annotation of historical cases is useful for benchmark construction but cannot substitute for prospective validation.

The ontology may be unstable

The shared transport ontology may not carve nature at its joints. What counts as a “failure mode” or a “feedback loop” may differ enough across domains to make cross-domain comparison spurious.

Response: Ontology instability is a known challenge in formal knowledge representation. The appropriate response is not to demand a perfect ontology before proceeding but to build the ontology incrementally, document disagreements, and measure the extent to which ontological choices affect contract outputs. The ontology is a hypothesis, not a fixed axiom.

Surface similarity may masquerade as structural equivalence

Layer 2 of the architecture may systematically confuse visually or linguistically similar representations for structurally equivalent ones, particularly if the contract discovery process relies on language model-based hypothesis generation.

Response: This is precisely why Layer 3 (Constraint Verification) exists. Contract discovery generates hypotheses; verification tests them. A system that generates high-quality hypotheses but verifies poorly is an incomplete research program. A system that generates poor hypotheses but verifies rigorously will still fail most of the time. Both layers must be developed in parallel.

Transport predictions may fail at scale

Even well-verified contracts may fail when the target system operates at scales, speeds, or complexity levels outside the scope of the verification. A contract verified on simple systems may not generalize.

Response: Scope conditions must be explicit in every contract. A contract is not a universal claim; it is a claim relative to a specified class of instances. Identifying the boundaries of that class is part of Contract Calibration and is itself a productive research outcome.

Expert disagreement may be irreducibly high

If domain experts cannot reliably agree on what is preserved versus not preserved, the framework may be less a science than a structured format for expert disagreement.

Response: High disagreement is informative. It identifies which preservation properties are genuinely contested, which domains are underspecified, and which contracts require more evidence. A framework that makes disagreements explicit and localized is more useful than one that allows them to remain implicit and diffuse.

10. Research Roadmap

Phase 1 — Ontology and Benchmark (Months 1–12)

Develop the Transport Ontology through structured workshops with experts across target domains. Construct the initial version of the Historical Transfer Benchmark with 30–50 annotated examples. Measure inter-rater reliability on property-level annotation. Deliverables: Transport Ontology v1.0; Benchmark Dataset v1.0; inter-rater reliability report.

Phase 2 — Contract Representation and Discovery (Months 6–18)

Develop the Process Representation Schema and Preservation Contract Schema. Implement Layer 0 and Layer 1 of the architecture. Develop and evaluate candidate contract discovery algorithms on benchmark cases. Deliverables: Schemas; Layer 0 and 1 implementations; contract discovery evaluation report.

Phase 3 — Human Agreement Studies (Months 12–24)

Design and conduct human studies with domain experts: can experts independently and reliably identify Preservation Contracts? Which properties are robustly identified, which are ambiguous, which are consistently missed? Deliverables: Human agreement study protocol; dataset; statistical analysis.

Phase 4 — Prototype Transport Engine (Months 18–36)

Implement Layers 2, 3, and 4. Integrate contract discovery, constraint verification, and predictive transport. Evaluate on held-out benchmark cases. Deliverables: Working prototype; evaluation against benchmark; error analysis.

Phase 5 — Prospective Validation (Months 30–48)

Generate novel transfer hypotheses prospectively. Collaborate with domain scientists to test transported predictions experimentally. Measure prediction transfer success rates. This phase is the strongest test of the framework and the most likely to produce surprises in both directions. Deliverables: Set of pre-registered transfer hypotheses; experimental results; revised framework.

Required expertise: knowledge representation, formal verification, machine learning, causal inference, information theory, domain scientists in biology, physics, and computer science, human-subjects research methodology.

11. Conclusion

Wavefront proposes to study transportability as a first-class scientific object. The primitive is the Preservation Contract: a formal, falsifiable specification of what survives transformation and what does not. The architecture has six layers, each with identified open research problems. The evaluation framework proposes concrete metrics against a historical benchmark and prospective experimental validation.

We do not claim that Wavefront will automate scientific discovery. We claim something more modest and more defensible: that humans already perform transport reasoning intuitively, that this reasoning is inconsistent and often unchecked, and that formalizing it makes it auditable, improvable, and scalable. Physics did not invent gravity. It formalized it. Information theory did not invent communication. It formalized it. Wavefront aims to formalize a class of reasoning

about structural translation that scientists and engineers have always performed but have never studied directly.

The most important output of a successful Wavefront program would not be a better analogy engine. It would be a calibrated science of when to trust an analogy and when to distrust it—and a method for generating new truths by identifying structural equivalences that no expert had previously recognized.

Do not ask: What is the thing? Ask: What survives the transformation? Find the mapping. Specify the contract. Test the transfer.

Notes on Scope and Connections

Connections to existing research programs: Category theory (Mac Lane; Awodey); causal representation learning (Schölkopf et al. 2021); mechanistic abstraction in AI interpretability (Geiger et al. 2023); minimum description length (Grünwald 2007); dynamical systems and control theory (Strogatz; Khalil); structural causal models (Pearl 2009); analogical reasoning in cognitive science (Gentner 1983; Holyoak and Thagard 1995).

Wavefront is distinguished from prior analogical reasoning systems (e.g., SME, Copycat) by its emphasis on falsifiability, explicit non-preservation claims, quantitative confidence, and prospective predictive validation. It is distinguished from pure formal methods by its tolerance for partial and uncertain specifications in complex empirical domains.